MECH321
Dynamics of Engineering Systems

Lecture 1
Introduction

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TYUT · 2026

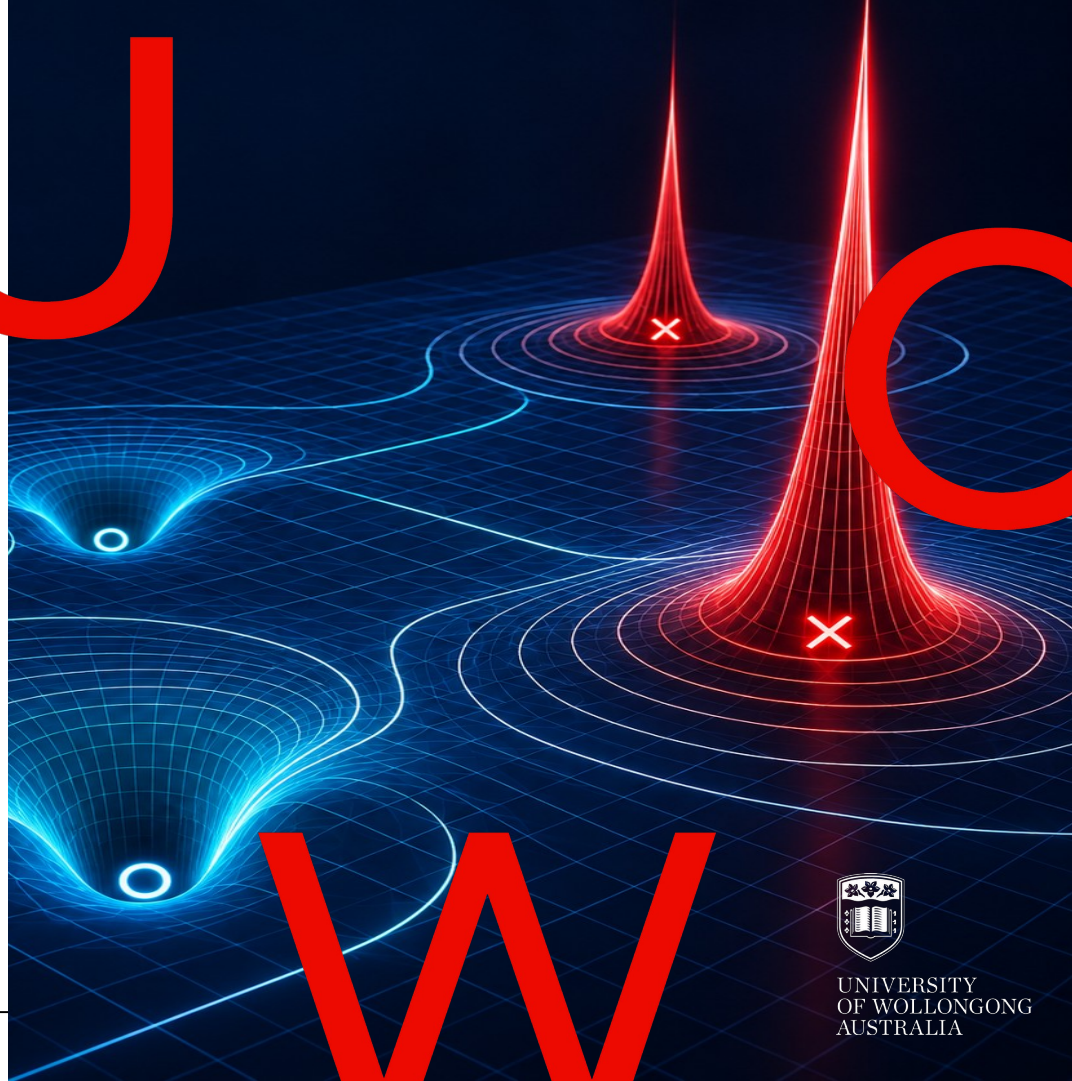


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Course Structure

CORE

Mathematical modelling

L1–L3, L6

L1: variables, states and model order

L2–3: ODE, Laplace and transfer function

L6: Physical Modelling and state model

Response analysis

L4–L5, L7–L8

L4: transient response and specifications (time domain)

L5: frequency response and validation (frequency domain)

L7: what feedback changes

L8: Nyquist, stability and margins

Control design

L9–L10 · OPTIONAL

L9: root locus;

L10: P, PI and PID

Physical and data-based models

White-box

Physical laws

Newton + spring + damper

$$m\ddot{x} + c\dot{x} + kx = f(t)$$

Black-box

Data alone

Neural network: input → output



This course uses white-box models based on physical laws.

Grey-box

Physics + data

Physics-informed learning

$$\Sigma F = m\ddot{x}$$ **+ observed data**

PART 1

State, storage and model order

Model order = number of independent stored quantities (state variables).

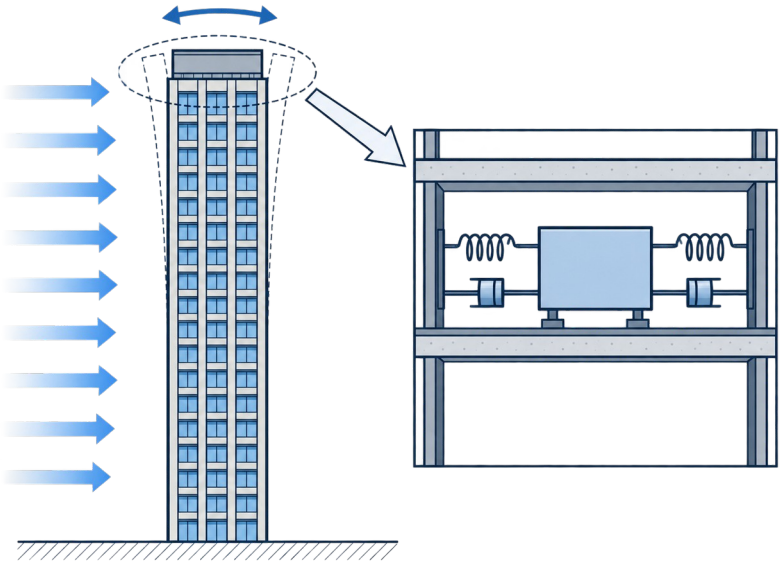
01 Static and dynamic response

02 Storage and state variables

03 Model order and applications

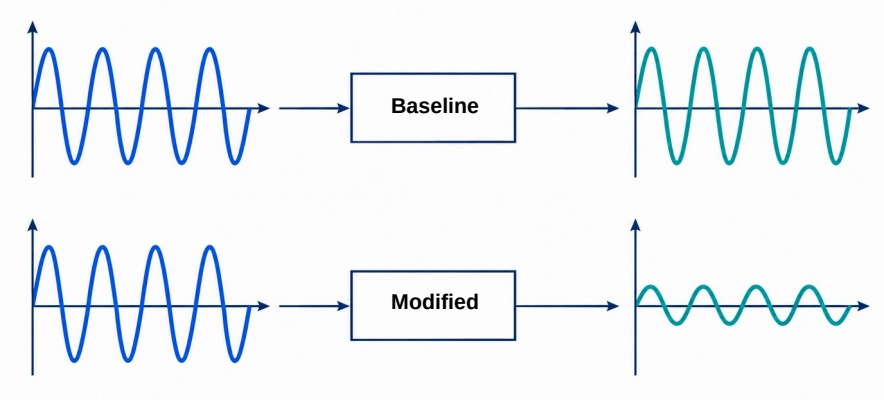
Skyscraper under wind loading

EXAMPLE



Schematic redrawn after Week 1 Lecture Notes (Rowell & Wormley, p. 4).

Same wind input → **different roof motion**



Input: wind force → Output: roof displacement or velocity

Model: mass and flexibility store energy; damping dissipates it.

At large sway, could the spring law become nonlinear?

Static and dynamic response

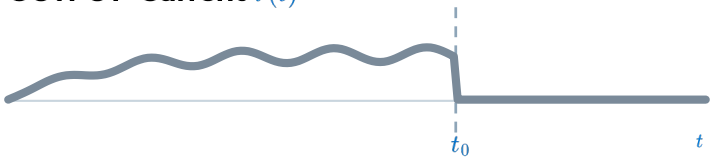
When a changing input ends, does the output end with it?

IDEAL RESISTOR

INPUT Voltage $v(t)$



OUTPUT Current $i(t)$



STATIC MODEL

Current follows the present voltage

No stored state.

FLEXIBLE BUILDING

INPUT Wind force $F(t)$



OUTPUT Roof displacement $x(t)$



DYNAMIC MODEL

Stored state affects future motion

Motion continues after the wind stops.

Concept check: memory

QUESTION

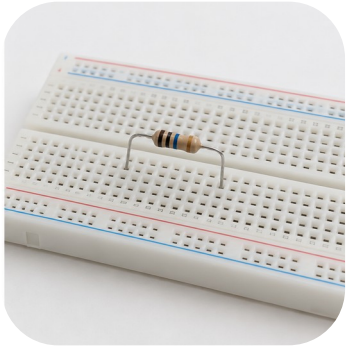
Which models keep an internal state after the input changes?



A Room + heater
Input: heater power
Output: room temperature



B Mass + spring
Input: applied force
Output: displacement



C Ideal resistor
Input: voltage
Output: current

Choose the dynamic models

A only · B only · A and B· A, B and C

Where the memory comes from

ANSWER

DYNAMIC

A and B retain stored energy

Room: thermal capacity stores internal energy

Moving mass: kinetic energy

Spring: elastic energy

STATIC IDEALISATION

C has no stored state

For an ideal resistor, $i\left(t\right) = \frac{v(t)}{R}$

Current follows the voltage at that instant

A real circuit has parasitics; whether to include them depends on the model.

State variables

Track the independent stored quantities.

Energy Storage and State Variables

Domain	Storage element and energy	State to retain	Balance law and loss	Typical model order
Mechanical	Mass: kinetic energy Spring: elastic potential	x and v for each independent moving mass	Newton's force balance; damper c dissipates energy	One moving mass: 2 states, second order
Fluid	Water tank: gravitational potential energy	Level h per tank	Volume balance; outlet resistance R limits outflow	One tank: first order
Thermal	Thermometer: internal energy	Temperature T of sensor	Energy balance; thermal resistance R_t limits heat flow	One sensor: first order
Electrical	Capacitor: electric field Inductor: magnetic field	Capacitor voltage v_C; inductor current i_L	KCL/KVL; resistor R dissipates electrical power	RC: first order; RLC: second order

Rule: count independent stored quantities, not the number of components.

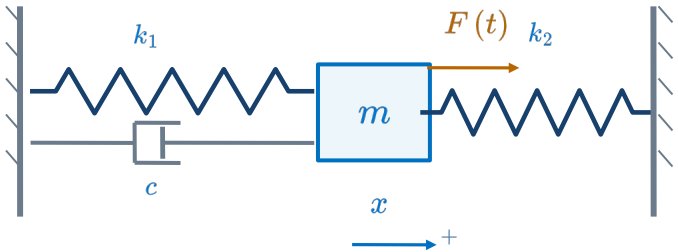
EXAMPLE

Which variables predict future motion?

For each model, find the minimum state variables and system order.

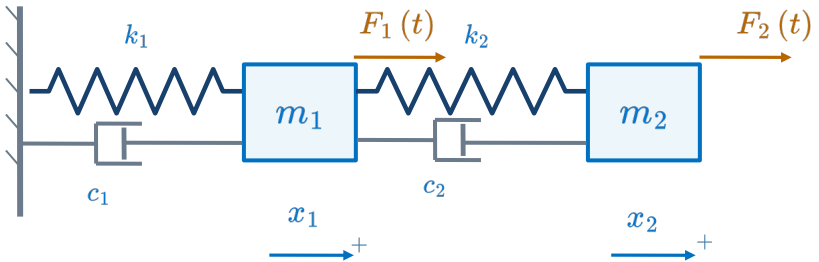
A

One mass between two wall springs, with damper c.
External force $F(t)$ acts rightward on m.



B

Two masses; m_1 connects to ground, m_2 to m_1.
External forces $F_1(t)$, $F_2(t)$ act rightward.



Applied forces are known inputs.

EXAMPLE

The equations explain the order

One second-order balance per independently moving mass.

A ONE MOVING MASS

Spring forces: $-k_1 x - k_2 x$

Damper force: $-c\frac{dx}{dt}$

$$m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + (k_1 + k_2)x = F$$

State: $\left[x, \frac{dx}{dt}\right]^T$

1 moving coordinate → 2 states → order 2

B TWO MOVING MASSES

Free-body forces are shown on the next slide.

$$m_1\frac{d^2x_1}{dt^2} + c_1\frac{dx_1}{dt} + k_1x_1 + k_2(x_1 - x_2) + c_2\left(\frac{dx_1}{dt} - \frac{dx_2}{dt}\right) = F_1$$

$$m_2\ddot{x}_2 + k_2(x_2 - x_1) + c_2(\dot{x}_2 - \dot{x}_1) = F_2$$

State: $\left[x_1, \frac{dx_1}{dt}, x_2, \frac{dx_2}{dt}\right]^T$

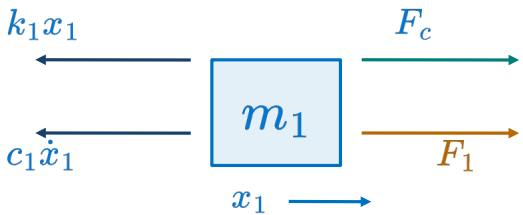
2 moving coordinates → 4 states → order 4

EXAMPLE

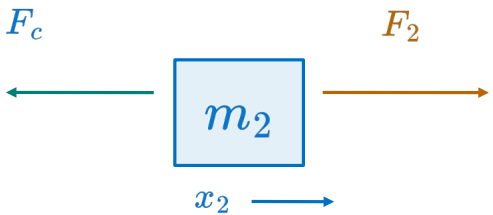
Free-body diagrams for the two masses

Applied forces F_1 and F_2 may be set to zero for free motion.

FORCES ON MASS 1



FORCES ON MASS 2



The interaction forces are **equal and opposite**. Negative values reverse the drawn arrows.

$$F_c = k_2\left(x_2 - x_1\right) + c_2\left(\dot{x}_2 - \dot{x}_1\right)$$

Newton's law on mass 1 (right positive)

$$F_1 + F_c - k_1x_1 - c_1\dot{x}_1 = m_1\ddot{x}_1$$

Newton's law on mass 2 (right positive)

$$F_2 - F_c = m_2\ddot{x}_2$$

Collect the balances to recover the equations on the preceding slide.

PART 2

Model classes and governing equations

01 Linearity and time dependence

02 Mechanical elements and worked examples

03 Cross-domain models and boundary choice

Linearity and time dependence in an ODE

CORE

LINEAR, TIME INVARIANT

EQX016001

$$\ddot{x} + 5\dot{x} + 10x = \sin t$$

sin t is the input; x **still enters linearly.**

LINEAR, CHANGING COEFFICIENT

EQX016002

$$\ddot{x} + [1 - \cos(2t)]x = 0$$

cos(2t) **changes a coefficient**, not the power of x.

NONLINEAR: x-DEPENDENT DAMPING

EQX016003

$$\ddot{x} + (x^2 - 1)\dot{x} + 10x = 0$$

x^2 multiplies $\dot{x}$; the unknown appears nonlinearly.

NONLINEAR FUNCTION OF x

EQX016004

$$\ddot{x} + \sin(x) = \cos t$$

sin(x) **acts on** x; cos t remains an input.

Nonlinearity acts on x: x^2, $x \cdot \dot{x}$ or sin(x).

A sin(t) input or cos($2t$) coefficient **still leaves a linear equation.**

Mechanical elements and equivalent stiffness

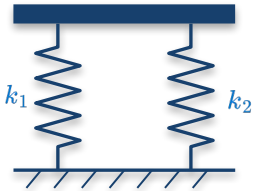
CORE

Mass m
$F = ma$
Resists acceleration.
Stores kinetic energy.

Damper b
$F = b\,(v_1 - v_2)$
Opposes relative velocity.
Dissipates mechanical energy.

Spring k
$F = k\,(x_1 - x_2)$
Opposes relative displacement.
Stores elastic energy.

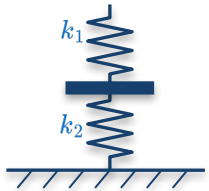
PARALLEL | same displacement



$$k_{eq} = k_1 + k_2$$

The two springs stretch by the same x; their forces add.

SERIES | same force



$$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$$

The same force passes through both; their extensions add.

Example 1: fixed-base mass–spring–damper

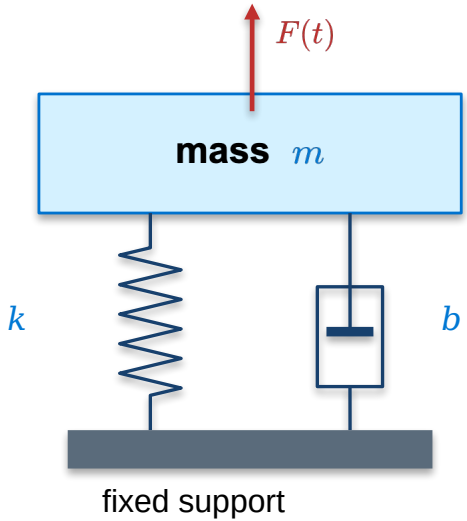
EXAMPLE

Problem

A 2 kg mass is attached to a fixed support by a spring of 80 N/m and a damper of 8 N·s/m.

At $t = 0$, apply a constant upward force of 2 N. Initially the mass is at rest at static equilibrium.

Take upward displacement $x\,(t)$ from that equilibrium. Derive the equation of motion and state the initial conditions.



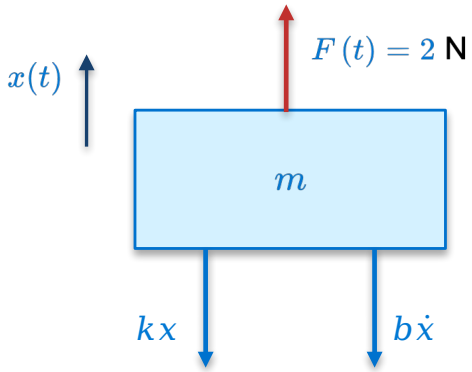
Hint: draw the mass alone

Mark the three forces on the mass. The support is fixed; using equilibrium displacement x cancels weight.

Example 1: free body to motion equation

EXAMPLE

The support is fixed; measure displacement from static equilibrium



1 ISOLATE THE MASS

Show the applied force upward. For a positive upward x and velocity, the spring and damper each exert a downward force.

2 READ THE ELEMENT FORCES

The lower ends do not move. Spring deformation equals x; relative damper velocity equals the velocity of the mass.

3 SUM FORCES

$$\Sigma F = F(t) - kx - b\dot{x} = m\ddot{x}$$

$$m\ddot{x} + b\dot{x} + kx = F(t)$$

EQX025001

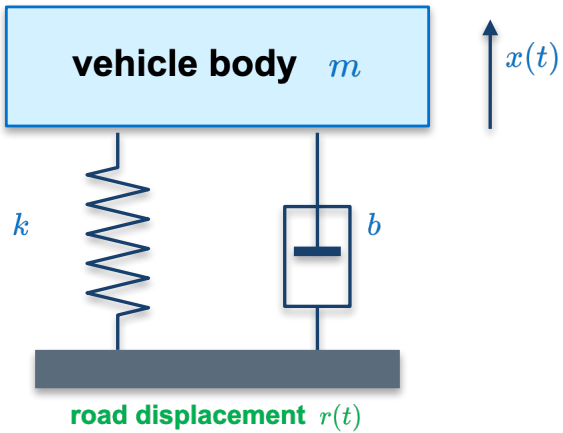
second order · linear for constant m, b and k · forced by $F(t)$

Substitute $m = 2$ kg, $b = 8$ N·s/m, $k = 80$ N/m and $F = 2$ N. Initially at rest:

$$2\ddot{x} + 8\dot{x} + 80x = 2, \quad x(0) = 0, \quad \dot{x}(0) = 0$$

Example 2: road-excited suspension

EXAMPLE



Problem

Use the same $m = 2$ kg, $k = 80$ N/m and $b = 8$ N·s/m as Example 1. Instead of an external force, the road rises at 25 mm/s for $0 < t < 1$ s.

Let $r(t)$ be road displacement and $x(t)$ body displacement, both positive upward from static equilibrium. The body starts from rest; the road starts at zero height.

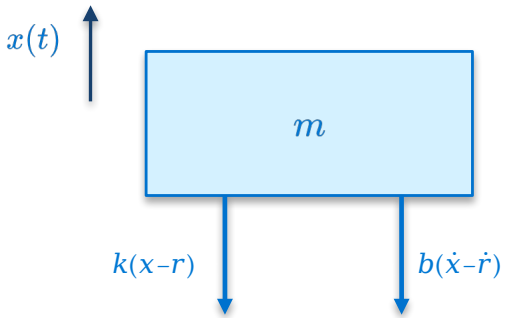
Derive the equation of motion. What numerical equation drives x during the ramp? Compare its input with the 2 N force case.

Hint: spring and damper forces depend on relative motion between the body and road.

Example 2: free body and suspension equation

EXAMPLE

Use relative displacement and velocity across the moving suspension



+x and +r point up; a negative relative term reverses its arrow.

Use the difference between the body and road motion at each element.

$$\Sigma F = -k\left(x-r\right) - b\left(\dot{x}-\dot{r}\right) = m\ddot{x}$$

$$m\ddot{x} + b\dot{x} + kx = b\dot{r} + kr$$

Road rise: 25 mm/s = 0.025 m/s. During the first second:

$$r\left(t\right) = 0.025t, \quad \dot{r}\left(t\right) = 0.025$$

0~ 1 s: $b\dot{r}\left(t\right) + kr\left(t\right) = 8 \times 0.025 + 80 \times 0.025t = 0.2 + 2t$

$$2\ddot{x} + 8\dot{x} + 80x = 0.2 + 2t$$

Example 1 and 2: same elements, different inputs

EXAMPLE

The support motion changes the force balance

Both systems: $m = 2$ kg, $b = 8$ N·s/m, $k = 80$ N/m; x and r are positive upward.

EXAMPLE 1 · APPLIED FORCE

$$2\ddot{x} + 8\dot{x} + 80x = 2$$

EXAMPLE 2 · MOVING ROAD

$$2\ddot{x} + 8\dot{x} + 80x = 0.2 + 2t$$

The left-hand dynamics are identical; only the input differs.

Both start at $x(0) = \dot{x}(0) = 0$. During $0 < t < 1$ s, $r(t) = 0.025t\ m$.

Series R–L–C circuit

Task: apply voltage $e_i(t)$; predict loop current $i(t)$ or capacitor voltage.

RESISTOR

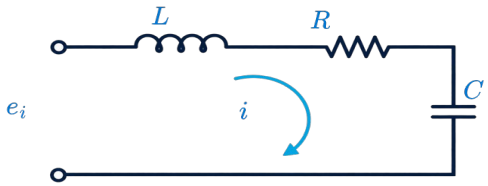
$$e_R = Ri$$

CAPACITOR

$$e_C = \frac{1}{C}\int i\,dt$$

INDUCTOR

$$e_L = L\frac{di}{dt}$$



KIRCHHOFF'S LOOP LAW

$$L\frac{di}{dt} + Ri + \frac{1}{C}\int i\,dt = e_i$$

HOW TO READ THE MODEL

The source voltage is shared among the three voltage drops. R **dissipates energy**; L **stores magnetic energy in current**, while C **stores electric energy in voltage**.

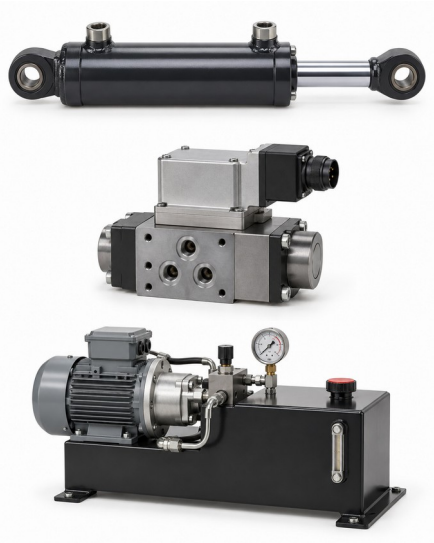
The two independent stores make this a **2nd-order system**. Current and capacitor voltage cannot jump instantaneously; differentiating the loop equation gives the familiar ODE.

See Lecture 6, slides 8–10: complex impedance method and mechanical–electrical analogy.

Hydraulic Servomotor Operation

OPTIONAL

The actuator is a piston and cylinder producing straight-line motion. A spool valve directs oil between the supply, cylinder chambers and return reservoir.



Top: cylinder. Middle: servo valve. Bottom: pump and reservoir. AI illustrations.

1 Supply and command. A pump supplies pressurised oil. The command moves the spool, which meters the oil. The pump supplies the power.

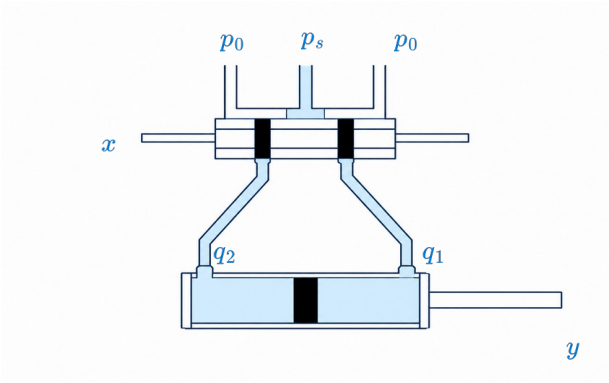
2 Flow and force. The valve connects one chamber to supply and the other to return. Chamber pressures exert a net force that moves the piston, rod and load.

3 Speed and reversal. Opening the valve further generally increases flow and speed. Reversing the spool swaps the supply and return paths, reversing motion.

4 Position control. A complete servo measures piston position and reduces the opening as the target is reached. An ideal closed-centre valve then blocks flow. Real feedback control corrects leakage and load changes.

Hydraulic servomotor

Task: position a load with hydraulic power. Valve displacement x meters flow from supply p_s to the cylinder; piston displacement y is the output.



IDEALISED FLOW MODEL

For small motions, valve mass flow q_1 is proportional to x. Treat the oil as incompressible and piston area A as constant.

$$q_1 = K_1 x$$

$$\rho A \frac{dy}{dt} = K_1 x$$

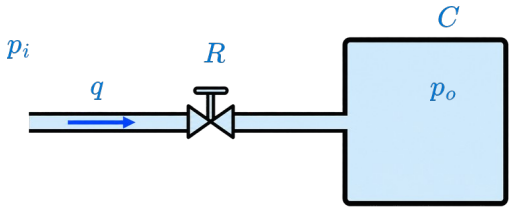
SYMBOLS, UNITS AND MODEL SCOPE

x, y [m]: valve-spool and piston displacement
p_s, p_0 [Pa]: supply and return pressure
q_1, q_2 [kg/s]: +y metered and opposite-port flows
ρ [kg/m³]; A [m²]: oil density; effective piston area

K_1 [kg/(s·m)]: local valve gain dq_1/dx
Continuity: moving by dy needs mass $\rho A dy$.
Fixed x gives nearly constant piston speed.
Omitted: leakage, compression, friction, load inertia.

Pneumatic pressure vessel

Predict vessel pressure p_o after inlet pressure p_i changes through the restrictive valve.



VALVE + AIR STORAGE

Near an operating point, R relates pressure drop to mass flow q. Pneumatic capacitance C is the air mass stored per unit pressure rise.

$$R = \frac{p_i - p_o}{q} \rightarrow \quad q = \frac{p_i - p_o}{R}$$

SYMBOLS AND UNITS

p_i, p_o [Pa]: inlet and vessel pressure
q [kg/s]: inlet air mass flow
R = (p_i−p_o)/q [Pa·s/kg]: valve resistance
C = dm/dp$_o$ [kg/Pa]: pneumatic capacitance
RC [s]: vessel filling time constant

FIRST-ORDER PRESSURE MODEL

$$RC\frac{dp_o}{dt} + p_o = p_i$$

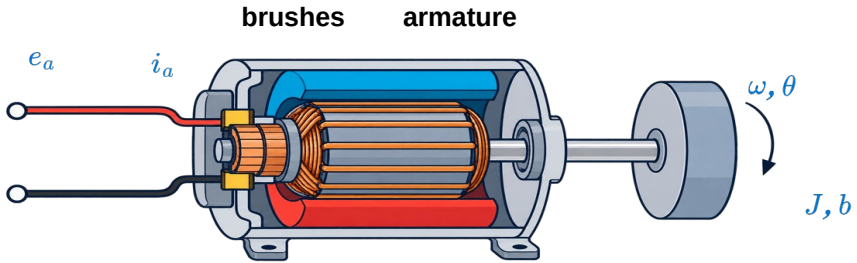
ASSUMPTIONS AND INTERPRETATION

Use small deviations near an operating point, with fixed volume and temperature. Mass balance makes p_o the stored state.

EXAMPLE

DC motor: electrical and mechanical dynamics

Predict shaft speed or position for armature voltage $e_a(t)$.



SYMBOLS AND UNITS

e_a, e_b [V]: applied voltage; back EMF
i_a [A]: current; R_a [Ω]: armature resistance
L_a [H]: inductance; θ [rad]: shaft angle
$\omega = \dot{\theta}$ [rad/s]: speed; J [kg·m²]: inertia
b [N·m·s/rad]: viscous friction coefficient
T [N·m]: torque; K [N·m/A]: torque gain
K_b [V·s/rad]: back-EMF gain

ARMATURE CIRCUIT

$$L_a \frac{di_a}{dt} + R_a i_a + e_b = e_a$$

SHAFT + BACK EMF

$$J\ddot{\theta} + b\dot{\theta} = T = Ki_a$$

$$e_b = K_b\dot{\theta}$$

States for speed: armature current i_a and shaft speed $\dot{\theta}$. Integrate speed for position θ.

Current produces torque; shaft motion creates opposing back EMF. Assume constant field and linear coefficients; external load torque is omitted.

$$L_a J\ddot{\omega}(t) + (L_a b + R_a J)\dot{\omega}(t) + (R_a b + KK_b)\omega(t) = Ke_a(t)$$

$$\omega(t) = \dot{\theta}(t)$$

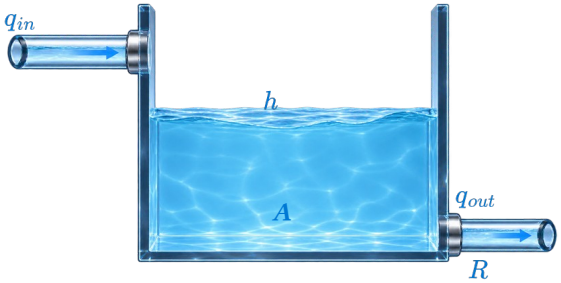
Voltage-to-speed model: a second-order system.

Water tank models

EXAMPLE

Find the level dynamics from volume balances and flow laws.

ONE TANK



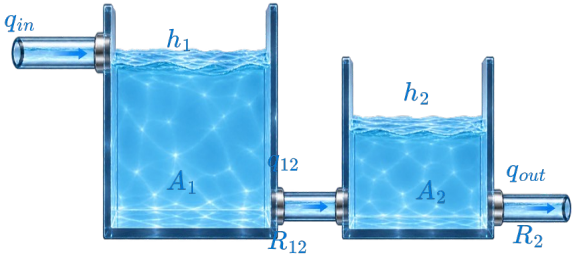
h [m]: level; A [m²]: tank cross-sectional area
q_{in}, q_{out} [m³/s]: inlet and outlet volume flow
$R = h/q_{out}$ [s/m²]: outlet head–flow resistance
One stored level gives a first-order model.

FLOW BALANCE + OUTLET LAW

$$A\frac{dh}{dt} = q_{in} - q_{out} \qquad q_{out} = \frac{h}{R}$$

$$A\frac{dh}{dt} + \frac{h}{R} = q_{in}$$

TWO CONNECTED TANKS



h_1, h_2 [m]: levels; A_1, A_2 [m²]: tank areas
q_{12}, q_{out} [m³/s]: intertank and outlet flow
$R_{12} = (h_1 - h_2)/q_{12}$; $R_2 = h_2/q_{out}$ [s/m²]
Two stored levels give a second-order model.

ONE BALANCE FOR EACH TANK

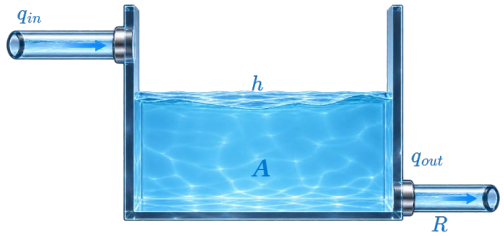
$$A_1\frac{dh_1}{dt} = q_{\text{in}} - q_{12}$$

$$A_2\frac{dh_2}{dt} = q_{12} - q_{\text{out}}$$

Where is the model boundary?

EXAMPLE

The water tank: one valve change, two modelling questions



Initially q_{in} = q_{out}. The outlet valve opens unexpectedly, reducing R while the pump keeps q_{in} fixed.

Symbol key (from slide 27): h [m] level; A [m²] area; q_{in}, q_{out} [m³/s] volume flows; R [s/m²] head–flow resistance. Changing R is a disturbance; fixed R is a parameter.

BOUNDARY A: VALVE ONLY

Input: *h*. Output: q_{out}.

$$q_{out} = \frac{h}{R}$$

The valve stores negligible water. With h and R known, q_{out} follows immediately.

BOUNDARY B: TANK + VALVE

Input: q_{in}. Output: *h*.

$$A\frac{dh}{dt} = q_{in} - q_{out}$$

The tank stores volume A h. Just after opening, q_{out} > q_{in}, so h falls gradually until flows balance.

$h\,(0)$ **[m] is the initial level. Boundary** B **needs it to predict the future; Boundary A does not.**

PART 3

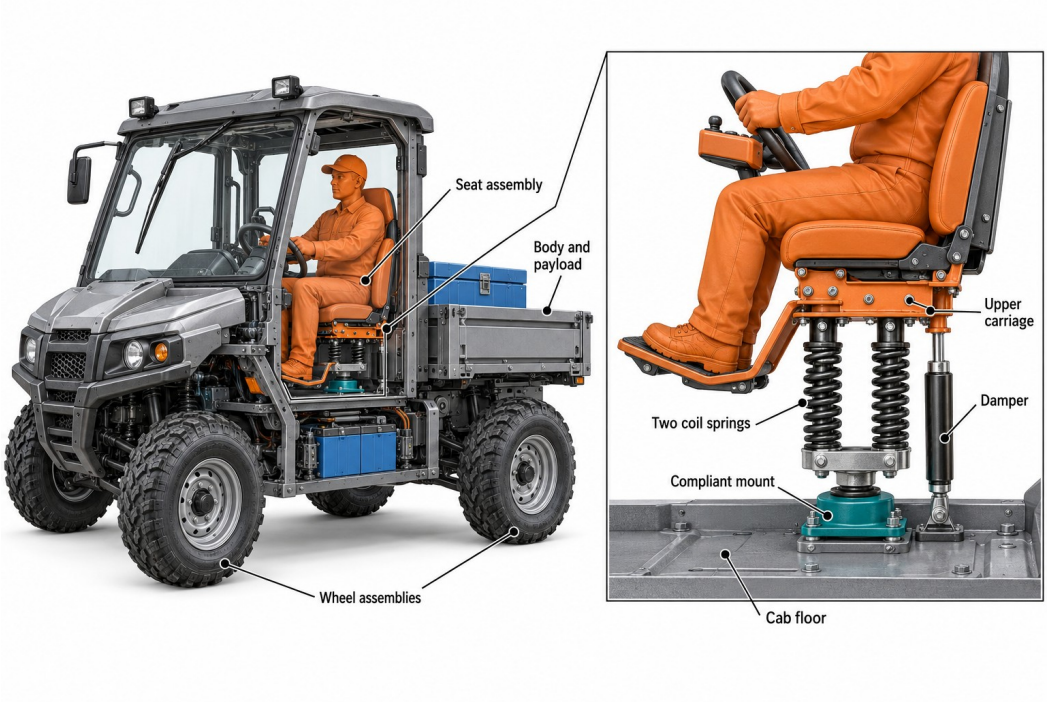
About the Projects

01 Project 1: vehicle seat suspension

02 Stage 1 in the Project 1 Word document

03 Project 2: robot joint speed control

Project 1: vehicle seat suspension



Physical model

Identify the moving mass.
Draw the equivalent model and free-body diagram.

Class discussion

Open Stage 1 in the Project 1 Word document.

Project 2: Robot joint speed control



Application photo: Robotiq. Project uses a separate joint motor.

WHAT IS MODELLED

A brushed DC motor drives a robot joint.
Armature voltage commands speed; an opposing load torque disturbs it.

PROJECT QUESTION

Can feedback hold 8 rad/s **within the ±24** V **limit?**

Test a 0.40 N·m opposing load step and check the required voltage.

Course tasks

Derive the voltage-to-speed model.
Evaluate P feedback: speed error, load offset and stability margins.
Check the voltage limit. State-space extension: optional.


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